

Supplementary Material

Bundle Neural Networks for Functional Connectomes

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1 Frozen analysis contract

The manuscript is compiled from the frozen Step 13 evidence package. The contract binds the claim ledger, result snapshot, artifact manifest, and paper asset manifest by SHA-256 digest. The manuscript-input builder refuses to generate tables or numeric macros if any bound input has changed. This supplement reports pre-specified secondary summaries; it introduces no new model fitting, participant exclusions, hyperparameter selection, or confirmatory inference.

The claim boundary is deliberately narrow. Results apply to the analyzed ABIDE-I C-PAC filtered, no-global-signal-regression, AAL-116 pipeline with positive-edge graphs and connectivity-profile node features. They do not establish clinical utility, anatomical or causal connectivity, biological bundle geometry, or universal inferiority of BuNNs.

2 Representation endpoints

Raw learned-BuNN embeddings can occupy node-specific frames. Cross-node diagnostics were therefore computed only after mapping embeddings into a common learned frame. The primary endpoint was normalized effective rank at layer 2, compared using each operator’s matched density-zero anchor. Dispersion, cosine similarity, invariant edge-transport distance, and encoder-to-layer-2 rank change were secondary diagnostics.

Table S1: BuNN-minus-GCN matched-anchor representation contrasts at layer 2.

Endpoint	Estimate	95% site-bootstrap CI
Normalized effective rank	-0.0072	[-0.0131, -0.0010]
Normalized dispersion	-0.0417	[-0.0523, -0.0306]
Mean pairwise cosine similarity	0.0415	[0.0319, 0.0507]
Invariant edge-transport distance	-0.6436	[-0.8205, -0.4720]
Encoder-to-layer-2 effective-rank change	-0.0056	[-0.0144, 0.0048]

3 Robustness summaries

The equal-site mean was the confirmatory aggregation because each held-out site defined one generalization unit. Participant-weighted and site-median summaries were pre-listed sensitivity analyses, not alternative routes to a positive conclusion.

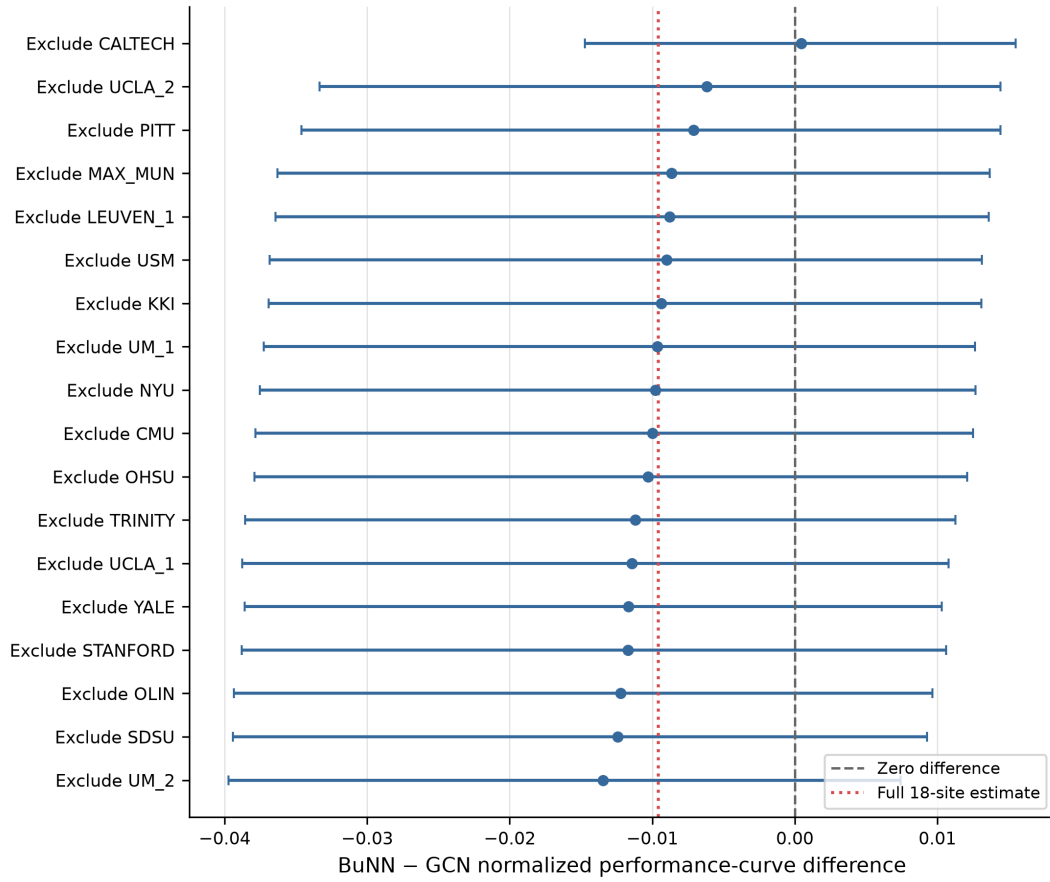


Figure S1: Leave-one-site-out influence analysis for the primary BuNN-minus-GCN curve contrast. Every pre-listed exclusion is shown.

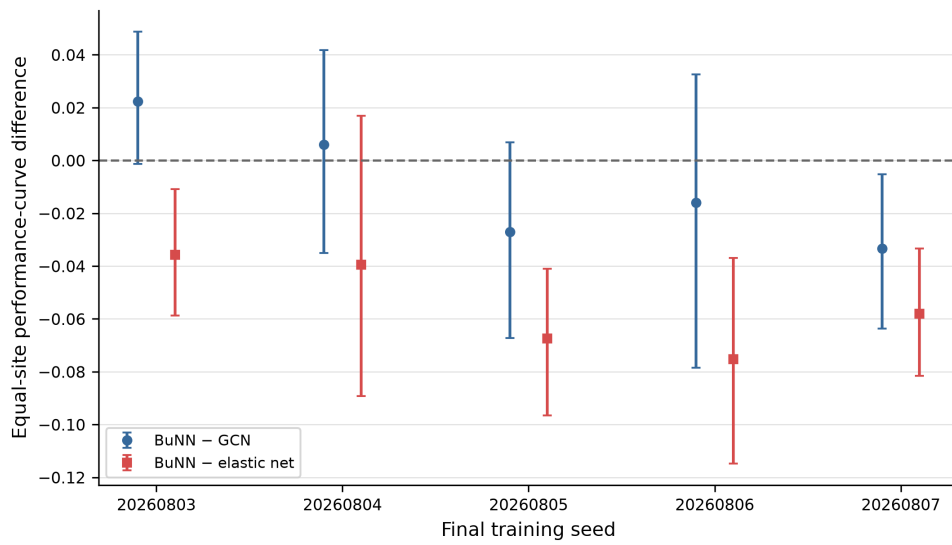


Figure S2: Seed-specific curve contrasts. Intervals resample held-out sites within each fixed final seed. Seed-level results are robustness diagnostics, not alternative primary analyses.

Table S2: Pre-listed alternative summaries. Equal-site means remain confirmatory.

Contrast	Summary	Estimate
BuNN - GCN curve	equal site mean	-0.00958
BuNN - GCN curve	participant weighted mean	0.00297
BuNN - GCN curve	median site difference	-0.00439
BuNN - elastic net	equal site mean	-0.05516
BuNN - elastic net	participant weighted mean	-0.04446
BuNN - elastic net	median site difference	-0.04644
BuNN - GCN effective-rank change	equal site mean	-0.00719
BuNN - GCN effective-rank change	participant weighted mean	-0.01094
BuNN - GCN effective-rank change	median site difference	-0.00917

The leave-one-site-out analysis recomputed each named contrast after excluding one outer site at a time. Seed-specific analyses recomputed the primary curve contrast separately for every fixed final seed. These analyses quantify site and optimization sensitivity; they do not replace the frozen primary estimand.

4 Reproducibility and integrity

The repository records the exact participant manifest, technical exclusion rules, connectome construction, held-out-site splits, operator contracts, tuning budgets, final seeds, source hashes, run audits, and aggregate result artifacts. Generated manuscript values are not copied manually: the build script reads the frozen result snapshot and writes LaTeX macros and tables. Unit tests verify the input hashes, deterministic generation, claim coverage, bibliography resolution, and the absence of manually duplicated primary results in narrative source files.